

Simplifying the PredRNN: A GRU-Based Approach

Project Report, Group 8, Applied Machine Learning

Augusta van Haren
s4809823

Hanna Hoogen
s1111075

Luca Pattavina
s1125484

Quincy Rommen
s1020956

June 20, 2025

Word count: 2041

1 Introduction

Recent developments in Artificial Intelligence (AI) have made the synthetic creation of images possible, with remarkable success (Betker et al., 2023). However, generating video frames remains a challenging endeavour, due to complex spatial-temporal dependencies within the data, and uncertainty in predicting future motion dynamics (Cai et al., 2018). This task, known as predictive learning, focuses on generating future data points conditioned on data points from the past. In the context of synthesis of video, the aim is to predict upcoming frames based on a sequence of past frames.

Tackling data of sequential nature was a key motivation for developing recurrent neural networks (RNNs). RNNs are a type of neural network that capture temporal dependencies within sequential data. Although RNNs have shown success capturing sequential dependencies within data, they suffer from a fundamental bottleneck: for long-term dependencies the recurrence becomes problematic, leading to the vanishing gradient problem when the input sequence is long (Hochreiter, 1998). This means that the network fundamentally forgets relevant information for its future predictions, decreasing the performance and accuracy of these predictions significantly.

To alleviate this problem, researchers developed two new components of RNNs: long short-term memory (LSTM) units, and gated recurrent units (GRUs). The LSTM consists of a cell with three gates: an input gate, an output gate and a forget gate. These gates keep track of relevant sequential features. This allows the network to keep track of temporal dependencies within the data, without the loss of information (Graves and Graves, 2012). The GRU is a later development and builds upon the concept of relevant information retention in a similar manner as LSTMs, but simplifies the architecture to have less parameters. The GRU consists of an update gate and a reset gate that modify the temporally dependent information flow (Chung et al., 2014).

Although LSTMs and GRUs have been used for speech or language modelling, an inherently non-Markovian process, researchers recently have developed an alternate form of LSTMs for predictive learning, specifically the future prediction of frames. For this type of predictive learning, LSTMs require both low-level details and high-level semantics for spatiotemporal data to generate future frames.

To achieve this, (Wang et al., 2017) developed the Predictive Recurrent Neural Network (PredRNN). PredRNN extends classic LSTMs by including a spatial-temporal memory flow in a special cell, to create the spatial-temporal LSTM (ST-LSTM). Spatial-temporal memory flow consists of bidirectional information processing hierarchies. Through this mechanism, top-down expectations and bottom-up signals are mixed to generate predictions of future input. This differs from classical LSTMs, where memory is preserved in each recurrent unit. Overall, the PredRNN showed better SSIM, PSNR and LPIPS than other models for predictions, indicating that the ST-LSTM was successful in accurately predicting future frames compared to previous models designed for the same task.

Considering the ST-LSTM improved performance on various datasets, combined with the knowledge that GRUs have less parameters to train, the current research focuses on the development of spatiotemporal GRUs (ST-GRUs). The prediction is that the ST-GRUs will be computationally less expensive and faster to train than the ST-LSTM. Additionally, faster inference time for the ST-GRU are also expected, due to less parameters, which may speed up the prediction process. The performance of ST-LSTMs and ST-GRUs will be compared and analysed to answer the following question: How do ST-GRU compare in performance to ST-LSTM in predictive learning for video prediction on the Moving MNIST dataset?

2 Methods

2.1 ST-LSTM-based PredRNN

PredRNN (Wang et al., 2017) consists of a stack of spatiotemporal LSTMs (ST-LSTMs) that employs a dual-memory mechanism, capable of learning the spatial correlations as well as temporal dynamics from sequential data – allowing it to accurately predict future spatiotemporal frames.

The ST-LSTMs in PredRNN handle four flows of information: the input state $\mathcal{X}_t$, hidden state $\mathcal{H}_t^l$, and *two* memory states – $\mathcal{C}_t^l$ and $\mathcal{M}_t^l$. These memory states are decoupled: $\mathcal{C}_t^l$ operates on a slower timescale and is designed to learn long-term dependencies in the data. It traverses horizontally from the previous node $t - 1$ to the current timestep t across the same layer. The so-called spatiotemporal memory state $\mathcal{M}_t^l$ instead, adapts more rapidly to changes in the input data, capturing short-term dynamics. It has a deeper transition path that zigzags in a bottom-up fashion, passing information from lower layer $l - 1$ to l at the same timestep – only the top layer of timestep $t - 1$ is latched to the current timestep t .

Both memory states are modulated by distinct sets of input i_t (resp. i_t'), forget f_t (resp. f_t'), and input modulation gates g_t (resp. g_t'). i_t determines how much of the new information from the current input state should be added – which is regulated by a sigmoid activation function (σ). Similarly, f_t decides what information should be discarded. Third, g_t generates candidate values that can be added to the memory state. Such candidate values are created using the hyperbolic tangent function ($\tanh$), which are then scaled by the input gate as $i_t \odot g_t$ and added to the memory state – replacing the forgotten information.

Finally, the updated memory states $\mathcal{C}_t^l$ and $\mathcal{M}_t^l$ are used to manipulate $\mathcal{H}_t^l$ – indirectly via output gate o_t , and directly via concatenation, 1×1 convolution and a $\tanh$ – which ultimately results in the new $\mathcal{H}_t^l$. The ST-LSTM architecture and its integration in PredRNN(-V2) is mathematically described in Figure 4a, and furthermore schematically visualised in Figure 5a and 6a of Appendix A.

2.2 ST-LSTM-based PredRNN-V2

PredRNN-V2 (Wang et al., 2021) further enhances this architecture. By introducing a new decoupling loss which explicitly extends the distance between the memory states in latent space – encouraging them to learn different aspects of spatiotemporal variations, and hereby preventing the learning of redundant features. Furthermore, it implements reverse scheduled sampling as a new curriculum learning strategy, which gradually changes the training process’s focus on using the previously generated frame to using the previous ground truth – enforcing learning the long-term dynamics. Lastly, it also enables action-conditioned video prediction.

2.3 ST-GRU-based PredRNN-V2

In this work, we present a variation on PredRNN-V2 that swaps the ST-LSTM building blocks for their GRU-counterparts – referred to as ST-GRUs. We designed this novel architecture ourselves, according to the key characteristics that define a GRU – modified to distinctly learn spatial and temporal features, as proposed by Wang et al. (2017, 2021).

In line with the GRU philosophy, we simplified the building block’s architecture by combining the memory updates in a simpler, unified form that only uses two gates: an update z_t (resp. z_t') and reset r_t (resp. r_t') gate. This design aims to reduce the parameter count and thus memory usage, hereby accelerating convergence when training PredRNN-V2 (Wang, 2023). Furthermore, it omits the short-term memory state $\mathcal{C}_t^l$ and instead, integrating its function in $\mathcal{H}_t^l$ – as they essentially follow the same horizontal flow from one timestep to the next.

This implies that the two gates manipulate $\mathcal{H}_t^l$ and $\mathcal{M}_t^l$ – instead of $\mathcal{C}_t^l$ and $\mathcal{M}_t^l$ in the ST-LSTM. z_t decides what part of the state is updated, whereas r_t defines what should be reset – very much in analogy with the i_t and f_t gate of the ST-LSTM (Chung et al., 2014). z_t and r_t are then used to render the candidate states $\tilde{\mathcal{H}}_t^l$ and $\tilde{\mathcal{M}}_t^l$ by updating and resetting the according information. These candidate states are then used to update $\mathcal{H}_t^l$ – indirectly via output gate o_t , and directly via concatenation, 1×1 convolution and a $\tanh$ – which finally results in the new $\mathcal{H}_t^l$. The ST-GRU architecture and its integration in PredRNN(-V2) is mathematically described in Figure 4b, and furthermore schematically visualised in Figure 5b and 6b of Appendix A – alongside with its ST-LSTM counterpart to stress their striking analogies regarding spatiotemporal memory decoupling, but at the same time highlighting their key differences in architectural features.

2.4 Experimental Setup

To evaluate the performance of the proposed version of ST-GRU-based PredRNN-V2, and directly compare it with the original ST-LSTM based version, we conducted the same experiment as Wang et al. (2021) on the Moving MNIST dataset, with the same training and testing practices. Specifically, we trained the model for

80,000 iterations using their training setup (Wu, 2022) and evaluate its performance on four different metrics: Mean Squared Error (MSE), Structural Similarity Index Measure (SSIM), Learned Perceptual Image Patch Similarity (LPIPS), Peak Signal to Noise Ratio (PSNR).

We further compare the performance of our model to that of other state-of-the-art models: ConvLSTM (i.e., a classical convolutional LSTM-based RNN *without* spatiotemporal memory decoupling) (Shi et al., 2015) and the ST-LSTM-based PredRNN (i.e., *with* spatiotemporal memory decoupling but *without* the V2-enhancements, see Section 2.2) (Wang et al., 2017). In addition, as we were developing our ST-GRU architecture, Dong et al. (2025) published their version of a ST-GRU, which they replaced as building blocks for the PredRNN architecture. As they tested its performance in complete agreement with the experimental procedure above, we also included their results in our comparison – as well as ConvGRU (i.e., a classical convolutional LSTM-based RNN *without* spatiotemporal memory decoupling) (Ballas et al., 2015). Regarding the ST-GRU-based PredRNN (Dong et al., 2025), we emphasize that this architecture differs from our ST-GRU-based PredRNN-V2 – *both* in terms of ST-GRU architectural details, as well as the presence of V2-enhancements (see Section 2.2).

3 Results

As shown in Figure 1, training the proposed ST-GRU-based PredRNN-V2 for 80,000 iterations resulted in a stable plateau in frame-wise MSE, SSIM, LPIPS, and PSNR metrics – indicating that the number of training iterations was sufficient for convergence. Furthermore, early frames are consistently predicted with higher frame-wise mean accuracy – an observation that is in line with our intuitions, since early frames are temporally closer to the input sequence, making them inherently easier to predict.

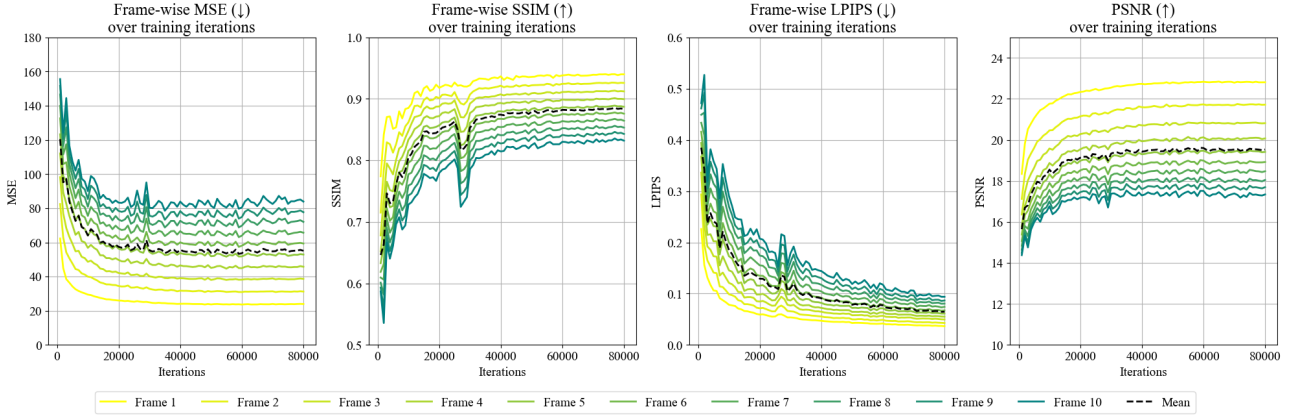


Figure 1: Frame-wise metrics (MSE, SSIM, LPIPS, PSNR) over 80,000 training iterations of the ST-GRU-based PredRNN-V2 on the Moving MNIST test set. The frame number indicates the timesteps between the predicted frame and last observed one.

Qualitative analysis of the model’s performance, based on the generated frames, confirm the higher predictive capabilities of the model for earlier frames (i.e., frames closer to the input frames) (see Figure 3). We can observe a gradual decline of local image features in late-sequence frames, especially in case of overlapping trajectories of the two represented digits.

3.1 Quantitative Model Comparison

Table 1 summarises the predictive performances of all aforementioned models – including the proposed ST-GRU-based PredRNN-V2 – in terms of MSE, SSIM, LPIPS and PSNR, averaged over the ten predicted frames. On both the frame-averaged MSE and SSIM, the ST-GRU-based PredRNN-V2 shows comparable performance to the ST-LSTM-based PredRNN – both vastly outperforming ConvLSTM and ConvGRU, but being inferior to the ST-GRU-based PredRNN, and ST-LSTM-based PredRNN-V2. In terms of LPIPS, the proposed ST-GRU-based PredRNN-V2 *does* outperform the ST-SLTM-based PredRNN-V2 – though the ST-GRU-based PredRNN again surpasses. But in terms of PSNR, the ST-LSTM-based PredRNN-V2 is again superior to replacing it with the proposed ST-GRU. Note that this last metric was obtained through direct evaluation of the ST-LSTM-based and ST-GRU-based PredRNN-V2 only.

The above-mentioned frame-averaged results typically generalise to the frame-wise model performances – as demonstrated by Figure 2. Here we again see comparable model performance of the ST-GRU-based PredRNN-V2 to the ST-LSTM-based PredRNN in terms of the frame-wise MSE and SSIM. The frame-wise LPIPS results were only available for the ST-LSTM-based models used for comparison – indicating superior performance of

Model	MSE ($\downarrow$)	SSIM ($\uparrow$)	LPIPS ($\downarrow$)	PSNR ($\uparrow$)
ConvLSTM (RNN) (Shi et al., 2015)	103.3	0.707	0.156	
ST-LSTM (PredRNN) (Wang et al., 2017)	56.8	0.867	0.107	
ST-LSTM (PredRNN-V2) (Wang et al., 2021)	48.4	0.891	0.071	20.047
ConvGRU (RNN) (Ballas et al., 2015)	57.8	0.839	0.091	
ST-GRU (PredRNN) (Dong et al., 2025)	53.4	0.893	0.057	
ST-GRU (PredRNN-V2)	55.1	0.883	0.065	19.529

Table 1: Model performance comparison on the Moving MNIST test set, reported in terms of MSE, SSIM, LPIPS, and PSNR. All metrics are averaged over ten predicted frames. The best performing model for each metric is highlighted in bold. Our model is highlighted in blue.

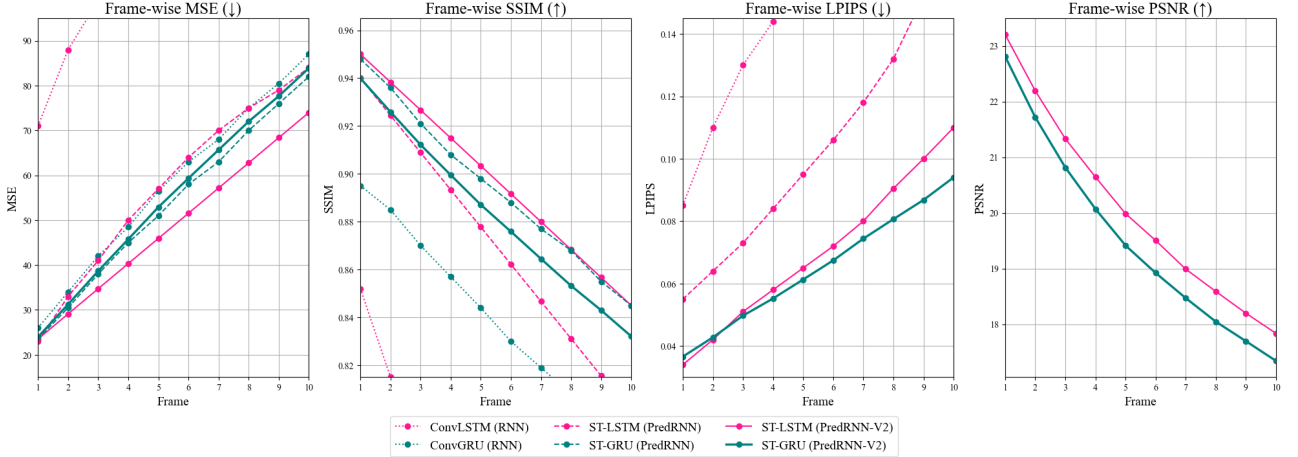


Figure 2: Model performance comparison on the Moving MNIST test set, reported in terms of frame-wise MSE, SSIM, LPIPS, and PSNR. Training and evaluation followed the procedure described in Section 2.4. Results for LSTM-based models are taken from Wang et al. (2021), and those for GRU-based models from Dong et al. (2025). PSNR for ST-LSTM-based PredRNN-V2 was computed via manual inference using the official, trained model (Wu, 2022). All metrics for the proposed ST-GRU-based PredRNN-V2 model were obtained through direct evaluation.

the proposed ST-GRU-based PredRNN-V2, especially on later frames in the sequence. The frame-wise PSNR was again in favour of the ST-LSTM-based PredRNN-V2 – in line with Table 1.

4 Discussion

We found that while our proposed ST-GRU-based PredRNN-V2 showed good performances across metrics and frames, it was generally outperformed by the original ST-LSTM-based PredRNN-V2. This is not surprising, considering the key improvements that define the ‘V2’. The newly introduced decoupling loss increases the distance between $\mathcal{C}_t^l$ and $\mathcal{M}_t^l$ – but as the ST-GRU does not operate on $\mathcal{C}_t^l$, this new feature has no benefit on the ST-GRU-based model. The experimental setup of this project did not involve action-conditioned video prediction, therefore, the ST-GRU-based PredRNN-V2 only benefitted from reversed schedule sampling – whereas the ST-LSTM-based PredRNN-V2 gained a larger advantage from both reversed schedule sampling as well as the new decoupling loss. Here, the proposed change to ST-GRU building blocks did not seem to outweigh those of the decoupling loss. However, it is interesting to note that the proposed model *did* score higher on the LPIPS – suggesting that the ST-GRU-basis allows to better capture spatiotemporal dynamics in terms of deeper, human-inspired features.

Furthermore, the finding that Dong et al. (2025)’s ST-GRU-based PredRNN implementation also outperformed the proposed ST-GRU-based implementation, confirms that their model architecture is more advanced for learning spatiotemporal dynamics. Given the much larger timescale they had for thorough model development and testing, this is understandable. Nevertheless, given the limited time of the current project, we are very satisfied with the achievements of the ST-GRU-based PredRNN-V2 – reaching similar predictive performances of ST-LSTM-PredRNN – but with a simplified building block architecture. We therefore encourage future research and development that builds upon the proposed model, refining architecture of ST-GRU and its integration in PredRNN-V2.

	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5	Frame 6	Frame 7	Frame 8	Frame 9	Frame 10
Input sequence										
True future sequence										
Predicted future sequence										
	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5	Frame 6	Frame 7	Frame 8	Frame 9	Frame 10
Input sequence										
True future sequence										
Predicted future sequence										
	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5	Frame 6	Frame 7	Frame 8	Frame 9	Frame 10
Input sequence										
True future sequence										
Predicted future sequence										
	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5	Frame 6	Frame 7	Frame 8	Frame 9	Frame 10
Input sequence										
True future sequence										
Predicted future sequence										
	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5	Frame 6	Frame 7	Frame 8	Frame 9	Frame 10
Input sequence										
True future sequence										
Predicted future sequence										

Figure 3: Five examples of model inference on five instances of the Moving MNIST test set – displaying the input sequence, true future sequence and predicted future sequence. By frame-wise comparison of the true vs. predicted future frames, we see that the similarity on both global as well as local features declines as predictions are further away in time.

5 Conclusion

In this work, we have presented the ST-GRU-based PredRNN-V2 – a reformulation of PredRNN-V2 that replaces its ST-LSTM building blocks by a newly-designed ST-GRU architecture – motivated by the reduced computational complexity of the ST-GRU that eases model training.

Evaluating the proposed architecture on the Moving MNIST showed that the model was successful in capturing the spatiotemporal dynamics of ten-frame future sequences – where predictive accuracy deteriorated for later frames in the sequence. Contrasting the ST-GRU-based PredRNN-V2 to comparable models showed similar, albeit inferior, performance to the ST-LSTM-based PredRNN (Wang et al., 2021) – but with less computational complexity.

References

- Ballas, N., Yao, L., Pal, C., and Courville, A. (2015). Delving deeper into convolutional networks for learning video representations.
- Betker, J., Goh, G., Jing, L., Brooks, T., Wang, J., Li, L., Ouyang, L., Zhuang, J., Lee, J., Guo, Y., et al. (2023). Improving image generation with better captions. *Computer Science*. <https://cdn.openai.com/papers/dall-e-3.pdf>, 2(3):8.
- Cai, H., Bai, C., Tai, Y.-W., and Tang, C.-K. (2018). Deep video generation, prediction and completion of human action sequences. In *Proceedings of the European Conference on Computer Vision (ECCV)*.
- Chung, J., Gulcehre, C., Cho, K., and Bengio, Y. (2014). Empirical evaluation of gated recurrent neural networks on sequence modeling.
- Dong, J., Zhang, J., Xie, B., Zhang, J., Liu, C., and Cheng, W. (2025). ST-GRU: spatiotemporal gated recurrent unit for video prediction. *Multimed. Syst.*, 31(3).
- Graves, A. and Graves, A. (2012). Long short-term memory. *Supervised sequence labelling with recurrent neural networks*, pages 37–45.
- Hochreiter, S. (1998). The vanishing gradient problem during learning recurrent neural nets and problem solutions. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, 6(02):107–116.
- Shi, X., Chen, Z., Wang, H., Yeung, D.-Y., Wong, W.-K., and Woo, W.-C. (2015). Convolutional LSTM network: A machine learning approach for precipitation nowcasting.
- Wang, Y. (2023). Advances in spatiotemporal graph neural network prediction research. *International Journal of Digital Earth*, 16(1):2034–2066.
- Wang, Y., Long, M., Wang, J., Gao, Z., and Yu, P. S. (2017). PredRNN: Recurrent neural networks for predictive learning using spatiotemporal LSTMs. In *Advances in Neural Information Processing Systems*, pages 879–888.
- Wang, Y., Wu, H., Zhang, J., Gao, Z., Wang, J., Yu, P. S., and Long, M. (2021). PredRNN: A recurrent neural network for spatiotemporal predictive learning.
- Wu, H. (2022). Predrnn: A recurrent neural network for spatiotemporal predictive learning (tpami 2022). <https://github.com/thuml/predrnn-pytorch>. Accessed: 2025-06-11.

A Model Architectures

A.1 Mathematical Comparison

$ \begin{aligned} g_t &= \tanh(W_{xg} * \mathcal{X}_t + W_{hg} * \mathcal{H}_{t-1}^l) \\ i_t &= \sigma(W_{xi} * \mathcal{X}_t + W_{hi} * \mathcal{H}_{t-1}^l) \\ f_t &= \sigma(W_{xf} * \mathcal{X}_t + W_{hf} * \mathcal{H}_{t-1}^l) \\ \mathcal{C}_t^l &= f_t \odot \mathcal{C}_{t-1}^l + i_t \odot g_t \\ \\ g'_t &= \tanh(W'_{xg} * \mathcal{X}_t + W'_{mg} * \mathcal{M}_{t-1}^{l-1}) \\ i'_t &= \sigma(W'_{xi} * \mathcal{X}_t + W'_{mi} * \mathcal{M}_{t-1}^{l-1}) \\ f'_t &= \sigma(W'_{xf} * \mathcal{X}_t + W'_{mf} * \mathcal{M}_{t-1}^{l-1}) \\ \mathcal{M}_t^l &= f'_t \odot \mathcal{M}_{t-1}^{l-1} + i'_t \odot g'_t \\ \\ o_t &= \sigma(W_{xo} * \mathcal{X}_t + W_{ho} * \mathcal{H}_{t-1}^l \\ &\quad + W_{co} * \mathcal{C}_t^l + W_{mo} * \mathcal{M}_t^l) \\ \mathcal{H}_t^l &= o_t \odot \tanh(W_{1 \times 1} * [\mathcal{C}_t^l, \mathcal{M}_t^l]) \end{aligned} $ (a) ST-LSTM	$ \begin{aligned} z_t &= \sigma(W_{xz} * \mathcal{X}_t + W_{hz} * \mathcal{H}_{t-1}^l + W_{mz} * \mathcal{M}_{t-1}^{l-1}) \\ r_t &= \sigma(W_{xr} * \mathcal{X}_t + W_{hr} * \mathcal{H}_{t-1}^l + W_{mr} * \mathcal{M}_{t-1}^{l-1}) \\ \widetilde{\mathcal{H}}_t^l &= \tanh(W_{xn} * \mathcal{X}_t + W_{rhn}(r_t \odot (W_{hn} * \mathcal{H}_{t-1}^l)) \\ &\quad + W_{mn} * \mathcal{M}_{t-1}^{l-1}) \\ \widetilde{\mathcal{H}}_t^l &= (1 - z_t) \odot \widetilde{\mathcal{H}}_t^l + z_t \odot \mathcal{H}_{t-1}^l \\ \\ z'_t &= \sigma(W'_{xz} * \mathcal{X}_t + W'_{mz} * \mathcal{M}_{t-1}^{l-1}) \\ r'_t &= \sigma(W'_{xr} * \mathcal{X}_t + W'_{mr} * \mathcal{M}_{t-1}^{l-1}) \\ \widetilde{\mathcal{M}}_t^l &= \tanh(W'_{xn} * \mathcal{X}_t + r'_t \odot (W'_{mn} * \mathcal{M}_{t-1}^{l-1})) \\ \mathcal{M}_t^l = \widetilde{\mathcal{M}}_t^l &= (1 - z'_t) \odot \widetilde{\mathcal{M}}_t^l + z'_t \odot \mathcal{M}_{t-1}^{l-1} \\ \\ o_t &= \sigma(W_{ho} * \widetilde{\mathcal{H}}_t^l + W_{mo} * \widetilde{\mathcal{M}}_t^l) \\ \mathcal{H}_t^l &= o_t \odot \tanh(W_{1 \times 1} * [\widetilde{\mathcal{H}}_t^l, \widetilde{\mathcal{M}}_t^l]) \end{aligned} $ (b) ST-GRU
--	--

Figure 4: Side-by-side mathematical comparison of ST-LSTM and ST-GRU building block architectures.

A.2 Visual Comparison

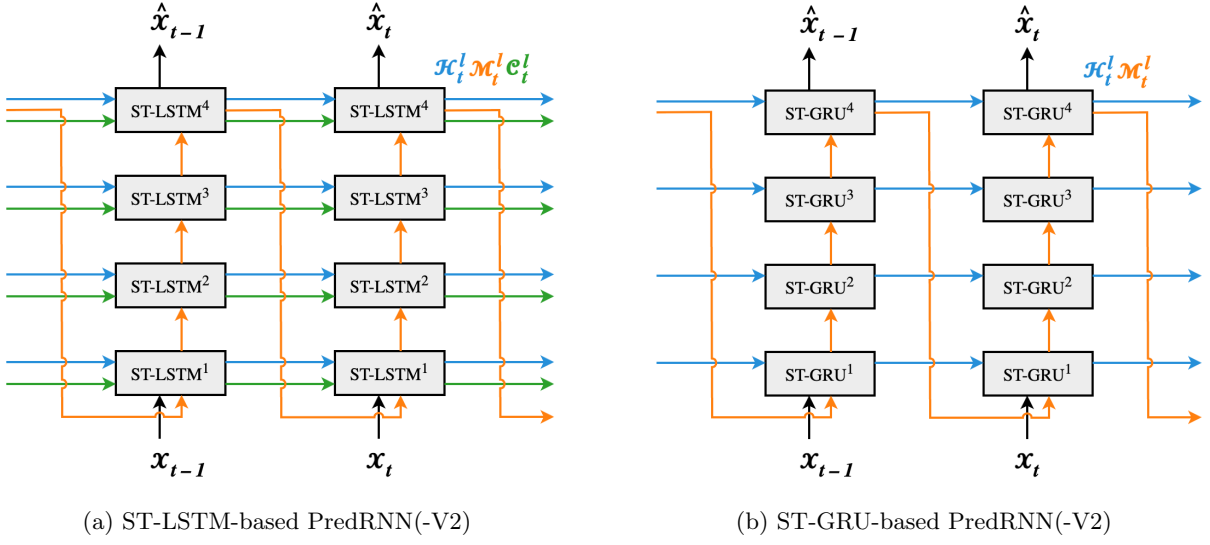


Figure 5: Side-by-side visual comparison of ST-LSTM-based PredRNN(-V2) and ST-GRU-based PredRNN(-V2) architectures – emphasising the spatiotemporal flow of information by using different colours. The ST-LSTM-based PredRNN(-V2) assigns two memory states – one for long-term memory $\mathcal{C}_t^l$ that flows horizontally from the previous timepoint to the current (t), whereas the spatiotemporal memory $\mathcal{M}_t^l$ zigzags through the levels (l) of PredRNN(-V2) – capturing short-term dynamics. Both memory states are used to update hidden state $\mathcal{H}_t^l$. The ST-GRU-based PredRNN(-V2), on the other hand, integrates the horizontal memory flow into the hidden state, by direct manipulation of $\mathcal{H}_t^l$. The spatiotemporal memory is still achieved by $\mathcal{M}_t^l$.

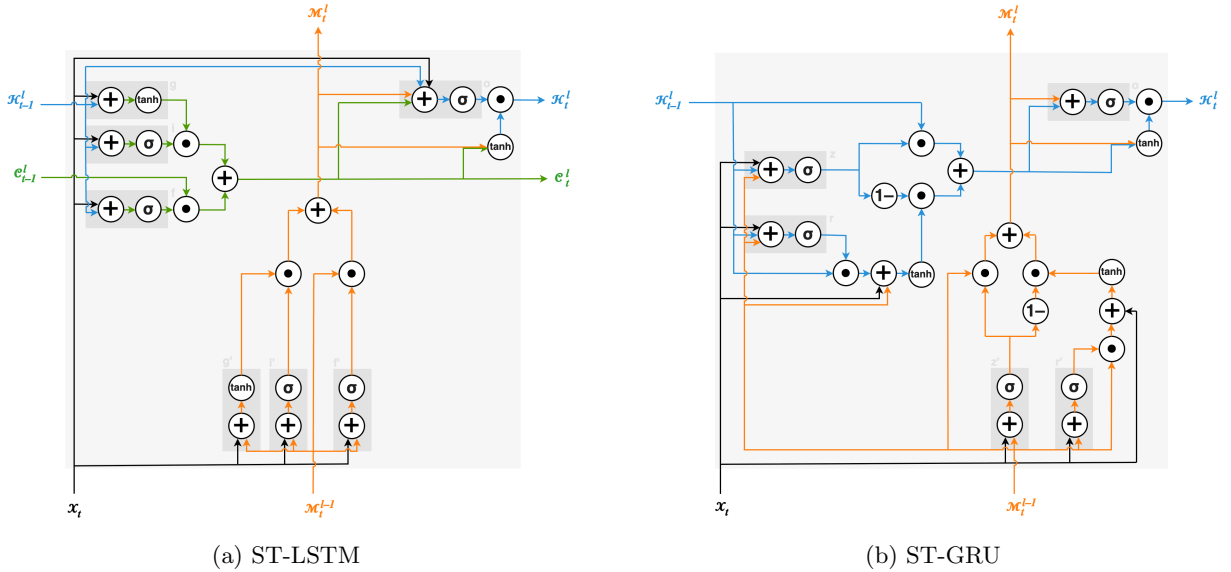


Figure 6: Side-by-side visual comparison of ST-LSTM and ST-GRU building block architectures – emphasising the spatiotemporal flow of information by using different colours. The ST-LSTM assigns two memory states – one for long-term memory $\mathcal{C}_t^l$ across timepoints (t), whereas the spatiotemporal memory $\mathcal{M}_t^l$ captures short-term dynamics across layers (l). Both memory states are individually modulated by three gates i_t , f_t and g_t (resp. i_t' , f_t' and g_t') – finally contributing to the updating of hidden state $\mathcal{H}_t^l$ via the output gate o_t . The ST-GRU, on the other hand, integrates the horizontal memory flow into the hidden state, by direct manipulation of $\mathcal{H}_t^l$. The spatiotemporal memory is still achieved by $\mathcal{M}_t^l$. Each is individually modulated by two gates z_t and r_t (resp. z_t' and r_t'), and finally combined via the output gate o_t . Both the reduction in gates as well as to-be-managed states are key to the ST-GRU's simplified architecture.